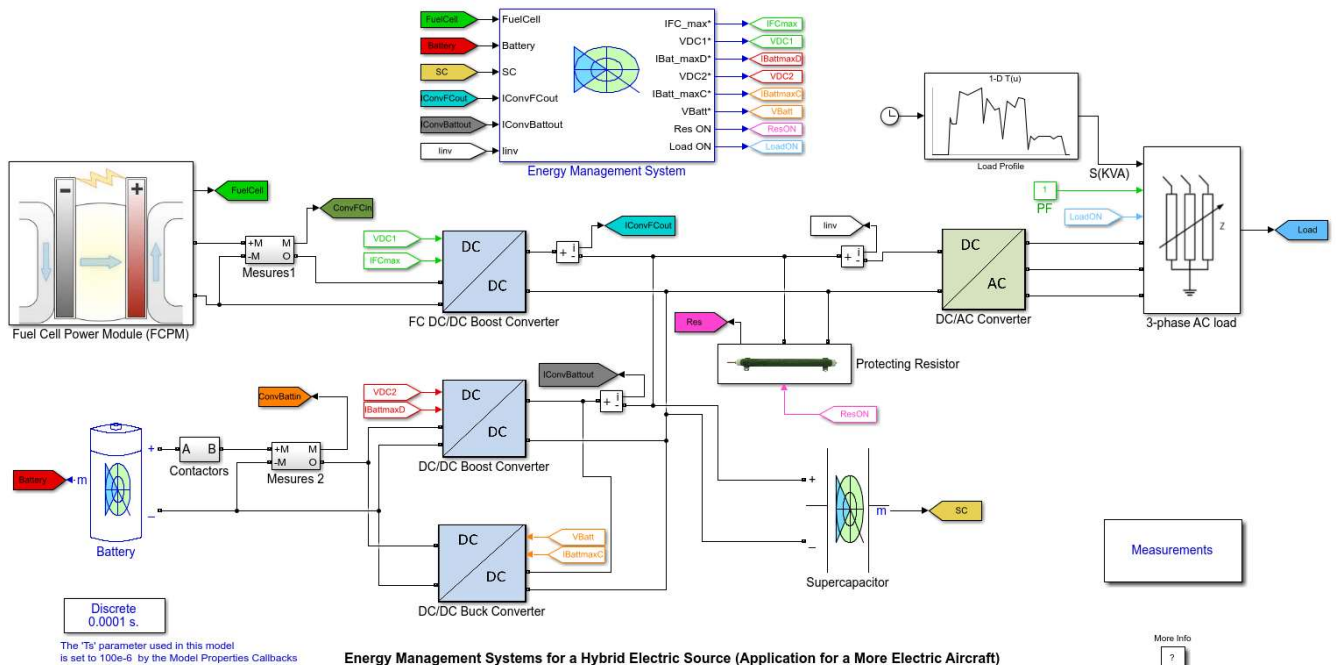


Energy Management Systems for a Hybrid Electric Source (Application for a More Electric Aircraft)

This example shows energy management systems for a fuel cell hybrid electric source.

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Souleman Njoya M., Louis-A. Dessaint (Ecole de technologie superieure, Montreal) and Susan Lisouet-Hanke (Bombardier Aerospace)



Circuit Description

This example illustrates a simulation model of a fuel cell based emergency power system of More Electric Aircraft (MEA). As the landing-gear and flight control systems become more electric in MEA, the peak electrical load seen by the conventional emergency power system (ram air turbine or air-driven generator) increases. Consequently, there is a potential risk of overloading the ram air turbine (RAT)/air-driven generator (ADG) at lower aircraft speeds, where the produced power is nearly zero. A more robust emergency power system is needed to ensure a safe landing of MEA. This model presents an alternative emergency power system based on fuel cells, lithium-ion batteries and supercapacitors. The demo also features different energy management systems for a fuel cell hybrid electric source.

The fuel cell hybrid power system is designed based on a representative emergency flight profile of a Bombardier aircraft and consists of the following:

- A 12.5 kW (peak), 30-60 V PEM (proton exchange membrane) fuel cell power module (FCPM), with nominal power of 10 kW.
- A 48 V, 40 Ah, Li-ion battery system.
- A 291.6 V, 15.6 F, supercapacitor system (six 48.6v cells in series)
- A 12.5 kW fuel cell DC/DC boost converter, with regulated output voltage and input current limitation.
- Two DC/DC converters for discharging (4 kW boost converter) and charging (1.2 kW buck converter) the battery system. These converters are also output voltage regulated with current limitation. Normally, a single bidirectional DC/DC converter can also be used to reduce the weight of the power system.
- A 15 kVA, 270 V DC in, 200 V AC, 400 Hz inverter system.

- A 3 phase AC load with variable apparent power and power factor, to emulate the MEA emergency load profile.
- A 15 kW protecting resistor to avoid overcharging the supercapacitor and battery systems.
- An energy management system, which distributes the power among the energy sources according to a given energy management strategy. Five types of energy management strategies are implemented, which are:
 1. The state machine control strategy
 2. The Classical PI control strategy
 3. The frequency decoupling and state machine control strategy
 4. The equivalent consumption minimization strategy (ECMS)
 5. The external energy maximization strategy (EEMS)

Demonstration

The demonstration shows the performance of the fuel cell hybrid emergency power system during a five minutes emergency landing scenario. In this scenario, the fuel cell hybrid power system supplies the essential loads during the following events:

- Instantly when the main generators are lost (this is normally assumed by the Avionic and APU battery system till the RAT/ADG is fully deployed).
- Emergency hydraulic pump start-ups.
- Motion of Flaps/Slats and gear down.
- Taxiing and passengers evacuation (also normally assumed by the Avionic and APU battery system as the RAT/ADG becomes unavailable).

Depending on the type of energy management strategy selected, the energy management system controls the power of each energy source devices through the reference signals (output voltage and maximum current) of the fuel cell and battery DC/DC converters. Double click on the **Energy Management System** block and select for example the **State Machine Control Strategy**. Start the simulation. Double click on the **Measurements** block. Open the **Power** scope (showing the power distribution referred to the 270 V dc bus) together with the **Fuel Cell**, **Battery**, **SuperCap** and **Load** scopes. The following explains what happens during this simulated emergency landing scenario:

At $t = 0$ s, the essential loads are supplied by the main generators and the fuel cell hybrid power system is turned ON to prepare for an unlikely emergency landing situation.

At $t = 5$ s, the fuel cell begins to recharge the battery with its optimal power (around 1 kW).

At $t = 40$ s, all generators are lost. The fuel cell hybrid power system takes over the essential loads. At this time the extra load power required is instantly supplied by the supercapacitor due to its fast dynamics, while the fuel cell power increases slowly.

At $t = 45$ s, the supercapacitor is discharged below the required DC bus voltage (270 V) and the battery starts providing power to regulate the bus voltage back to 270 V.

At $t = 48.5$ s, the DC bus or supercapacitor voltage reaches 270 V and the battery reduces its power slowly to zero. The fuel cell provides the total load power and continues to recharge the supercapacitor.

At $t = 60$ s, an emergency hydraulic pump is started, and the supercapacitor provides the extra transient load power, while the fuel cell power increases slowly.

At $t = 61.5$ s, the battery comes online to regulate the DC bus voltage to 270 V and helps the fuel cell by providing the extra load power required.

At $t = 70$ s, the fuel cell reaches its maximum power (the FCPM power was limited to 9 kW due to its DC/DC converter input voltage range) and the extra load power is provided by the battery.

At $t = 110$ s, the battery also reaches its maximum power (4 kW) and the supercapacitor provides the extra load power.

At $t = 125$ s, the load power reduces below the fuel cell maximum power. Due to the slow fuel cell dynamics, the extra fuel cell power during transients is transferred to the supercapacitor.

At $t = 126$ s, the DC bus voltage reaches 270 V and the battery power drops to zero.

At $t = 130$ s, a second emergency hydraulic pump is turned ON and the fuel cell hybrid power system behavior is similar to when the first hydraulic pump was turned ON.

At $t = 170$ s, the load power reduces below the fuel cell maximum power and the extra fuel cell power is transferred to both the battery and supercapacitor.

At $t = 180$ s, the load is suddenly increased due to the motion of Flaps/Slats and landing gears. Once again, the supercapacitor responds quickly by providing the extra load power.

At $t = 185$ s, the battery discharges to regulate the DC bus voltage and helps the fuel cell with the extra load power required.

At $t = 235$ s, the aircraft has landed and the load power decreases suddenly. The extra fuel cell energy is stored in the battery and supercapacitor.

At $t = 250$ s, the aircraft is taxiing and the fuel cell supplies nearly the total load power required.

At $t = 330$ s, the passengers have been evacuated and the load power reduces to zero. The fuel cell reduces its power slowly to its optimal power and recharges the battery.

Notes

1. In order to reduce the amount of memory used, a decimation factor of 100 is used for all scopes, except for the **Load** scope (which uses a decimation factor of 10).
2. Average-value models of DC/DC and DC/AC converters are used to speed up the simulation.
3. Select a different energy management strategy in the **Energy Management System** block and compare its performance in terms of hydrogen consumption, storage (battery/supercapacitor) energy used and overall efficiency.

References

1. S. Njoya Motapon, L.A. Dessaint and K. Al-Haddad, "A Comparative Study of Energy Management Schemes for a Fuel Cell Hybrid Emergency Power System of More Electric Aircraft," IEEE Transactions on Industrial Electronics, 2013 (IEEE Early access).